Radau: Numerical Method and Adaptive Execution

1 Differential equation and numerical method

Consider $z' = f(t, z)$, $z(t_0) = z_0$, in the project's packed physical layout. A three-stage Radau IIA collocation method of order five, with an embedded third-order error estimator and a cubic collocation interpolant. The implementation delegates the numerical tableau, local error estimator, trial rejection and interpolation polynomial to the installed SciPy `Radau` solver. The adapter does not duplicate these algorithms.

A Runge–Kutta evaluation has stage equations

$$Y_i = z_n + h \sum_j a_{ij} f(t_n + c_j h, Y_j), \qquad z_{n+1} = z_n + h \sum_i b_i f(t_n + c_i h, Y_i).$$

The collocation equations are implicit and are solved by SciPy Newton iterations. The public adapter accepts an optional physical-state Jacobian callable; otherwise SciPy uses finite differences. Its dimension is unchanged by energy tracking.

2 Error control and accepted intervals

The adaptive backend scales local error estimates using positive absolute and relative tolerances, with component scale based on $\mathrm{atol} + \mathrm{rtol} \max(|z_n|, |z_{n+1}|)$. It selects subsequent step sizes and can reject trial steps. `max_step` bounds accepted duration; `first_step=None` delegates initial selection to the backend. Requested tolerances bound neither global trajectory error nor roundoff.

One live solver is retained over the complete interval. Its state, stage work and any Jacobian/factorization reuse survive accepted steps. The adapter calls `solver.step()` until acceptance or failure; rejected trials never become observer records or separate accepted metric rows.

3 Dense output and energy extension

One retained SciPy implicit adaptive solver. Energy tracking is optional and passive. Eight-point Gauss quadrature integrates the passive derivative along each accepted dense interpolant. Partial-interval quadrature supplies off-grid energy samples. The user Jacobian covers physical coordinates only. Refine the physical step to audit the diagnostic quadrature, whose error is not controlled separately. The common state conventions are specified below.

4 Shared architecture and observation

The public class owns initialization, advance, observation and history export. Inherited `new_run` constructs a fresh instance of the same class and calls `initialize`, which creates its live solver. There is no separate method context or record of callbacks. All 13 methods inherit the same integration entry point and use the common global coordinator.

`ScipyAdaptiveController` exposes accepted intervals from the retained numerical session. The common collector stores requested samples and small accepted-step statistics; it does not retain interpolants or events by default.

An `AdaptiveIntegrationStep` records times and copied states. It also exposes work counters and `dense_state`. It does not claim a fixed-duration candidate-state map: adaptive decisions depend on numerical history. `AdaptiveTrajectoryObserver` explicitly retains continuous output outside the integrator. The observer and any persistence belong to the caller.

Function evaluations, Jacobian evaluations and LU decompositions are counted per accepted interval. Initialization work belongs to the first row; rejected trial work belongs to the acceptance that follows it. Interpolation needed only for an observer is excluded from main counters unless continuous output was explicitly requested. The ordinary method operation `advance` accepts one step on the retained solver, taking an upper step bound. The controller reports the actual accepted endpoint. A run instance belongs to one execution; repeating `simulate` creates a fresh instance and solver.

5 Implementation and verification

The adapter is in `src/methods/adaptive/scipy.py`; shared coordination and collection are in `src/execution/_host.py`. The current state and execution contracts are documented in its `simulation/` directory. Tests compare sampled states and work counters with the same SciPy solver, check a known non-autonomous rotation, and verify optional energy, dense retention, repeated runs, observation isolation and failure handling. Scientific reference trajectories still require refinement and an independent audit; arithmetic precision and integration error remain separate.

6 Four state formulations and passive energy tracking

For one planar particle the supported internal representations are

physical	$z = (x, y)$	$\mathbb{R}^2$,
physical with energy	(z, t, κ)	$\mathbb{R}^4$,
duplicated	$(u, v), \quad u, v \in \mathbb{R}^2$	$\mathbb{R}^4$,
duplicated with energy	(u, v, t, κ)	$\mathbb{R}^6$.

Classical methods use the physical rows; ABBA and BM4 use the duplicated rows. `track_energy` selects the energy variant. For N planar particles the respective dimensions are $2N$, $4N$, $4N$, and $6N$: one time entry and one momentum per particle. All time entries equal the integration time. Classical full-cyclotron runs instead use $4N$ or $6N$ entries. Accepted spatial copies coincide; only one physical copy is exported. Every coordinate block contains N values in component-major order. The formulation validates particle times and aligns output round-off; the integrator does not interpret coordinate indices.

The normalized passive balance is

$$\dot{\kappa} = -\partial_t H(t, z), \qquad \kappa(t_0) = 0, \qquad \Delta K(t) = H(t, z(t)) + \kappa(t) - H(t_0, z_0).$$

Splitting accumulators from both copies are divided by two. The physical Hamiltonian may vary with time; ΔK measures its computed balance. Tracking leaves physical trajectories, nonlinear stopping rules and adaptive acceptance unchanged. Dynamics must supply the momentum derivative, including zero for an autonomous Hamiltonian.

Hairer's constraint is solely $u - v = 0$: the reduced multiplier has $2N$ entries and ABBA's simultaneous spatial solve has $6N$ unknowns. Neither t nor κ is projected. The former eight-component projection is retired; `state_extension='fully_extended'` raises migration guidance.

Dynamics, formulation, method and integration remain separate. The shared state classes own layouts, diagonal embedding and history extraction in

```
src/formulations/state.py
```

Each integration initializes a fresh method instance. Observers receive physical states; energy histories are in `Solution.diagnostics`. The `extended_time` history has shape (N, n_{samples}). Fixed-step shadow samples leave main states and counters unchanged. Adaptive energy quadrature uses accepted dense output; its backend state and Jacobian remain physical.

7 Optional JAX execution

The optional JAX execution choice preserves the same adaptive method and SciPy controller, including accepted/rejected step control and dense output. JAX evaluates particle field and passive-energy batches on a selected CPU or GPU, returning them to the host solver. This is explicitly reported as hybrid execution; it does not move the adaptive solver or Radau linear algebra onto the GPU. Physical error norms and passive energy decoupling are unchanged.

Both execution routes use double precision and return the same physical solution layout. Floating-point reassociation can produce round-off differences without changing the designed order or geometric properties. First-call compilation is separate from warmed integration timings, which include transfers and output construction. The sibling runtime document and shared execution contract specify supported callbacks and exact device scope.